

ACCTG 211H — Mechanics Drill Set

Week 8 — Long-Term Assets, Depreciation, and Impairment

Drill Set

Required: 1, 2, 5, 6, 7, 8, 9. Due Sunday March 7, 2027, 11:59 p.m. Eastern, after both class meetings. Graded for completion: attempt the required items yourself, check the key, and correct your work. No generative AI. Optional items: 3–4; optional items do not count toward completion or required assessment.

1. (*Capitalized cost.*) A firm buys a machine: invoice price \$50,000, freight-in \$2,000, installation \$3,000, and prepaid insurance for the coming year \$1,500. Compute the cost to be capitalized.
2. (*Straight-line depreciation.*) The machine costs \$55,000, has a \$5,000 salvage value, and a five-year useful life. Compute annual depreciation and the book value at the end of year 2.
3. **Optional.** (*Double-declining-balance.*) For the same machine (cost \$55,000, salvage \$5,000, five-year life), compute depreciation expense for years 1 and 2 under the double-declining-balance method.
4. **Optional.** (*Units-of-production.*) A machine costs \$80,000, has an \$8,000 salvage value, and is expected to produce 180,000 units over its life. It produces 30,000 units in year 1. Compute year-1 depreciation.
5. (*Partial-year depreciation.*) Equipment costing \$24,000 (no salvage, four-year life, straight-line) is purchased and ready for use on October 1. Compute depreciation for the first partial calendar year.
6. (*Disposal.*) Equipment with a cost of \$40,000 and accumulated depreciation of \$30,000 is sold for \$7,000. Compute the gain or loss and record the journal entry.
7. (*Impairment.*) Treat a held-and-used long-lived asset as the entire relevant asset group for this simplified U.S. GAAP case. First compare carrying amount with undiscounted cash flows; if not recoverable, write down to fair value. It has a book value of \$120,000, estimated undiscounted future cash flows of \$110,000, and a fair value of \$95,000. Is the asset impaired, and if so, by how much?
8. (*Goodwill.*) In a simplified acquisition of an entire business (no prior ownership or noncontrolling interest), an acquirer pays \$500,000 for a target whose identifiable net assets have a fair value of \$430,000. Compute goodwill.
9. (*Intangible amortization.*) A patent is acquired for \$60,000 and has a remaining useful life of ten years, no residual value, and straight-line amortization. Compute annual amortization and the carrying value at the end of year 3.